# The Forced Derivation of $K_4$ from Logical Distinction

A Machine-Verified Chain in Agda, Reached by Two Independent Paths

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**Abstract**

The complete graph $K_4$ is the unique fixpoint of a self-consistency filter starting from a single axiom: the existence of a distinction. This result is established by a machine-verified development in Agda under `--safe --without-K` with zero postulates.

*Void* [1] (∼25 000 lines) derives $K_4$ by elimination: starting from abstract type-theoretic infrastructure—propositional equality, universe lifting, contractibility, distinct-pair classification—it destroys every alternative that fails self-consistency until only $K_4$ remains. The invariant record $(V, E, d, \chi, \kappa) = (4, 6, 3, 2, 8)$ with Laplacian spectrum $\{0, 4, 4, 4\}$ is proved to be a singleton.

A companion volume, *Form* [2] (∼9 000 lines), imports *Void*'s results and identifies the $K_4$ invariants with measured physical constants. The identification layer adds no new axioms; it maps the forced arithmetic onto the Standard Model's numerical structure. This paper presents the formal chain inside *Void* that produces $K_4$.

## 1 Introduction

Given the weakest possible starting point—a type with exactly two distinguishable elements—what combinatorial structure does self-consistency force?

The answer is $K_4$, the complete graph on four vertices. *Void* derives this result by elimination: every alternative is tested for self-consistency, and those that fail are destroyed. What remains is not chosen but forced. A companion volume, *Form*, imports *Void*'s definitions and maps the invariants onto measured physics.

The logical spine of the eliminative chain has five links.

1. Distinction forces four endomorphisms.
2. Level iteration forces a well-founded drift hierarchy.
3. Acyclicity forces every drift chain to terminate.
4. Observables collapse to a canonical presentation fixpoint.
5. The fixpoint forces $K_4$ as the unique surviving graph.

Each link is machine-checked in *Void*. This paper presents the formal chain itself, independent of the physical identifications (which are treated in the companion physics paper [3]).

**Related work.** Exhaustive classification of finite structures under algebraic constraints has a long history (Cayley's enumeration of groups, Burnside's lemma, modern computational algebra). The novelty here is not classification *per se* but the demonstration that a single binary distinction, iterated through universe levels, forces exactly one graph through purely type-theoretic means. The work closest in spirit is the programme of homotopy type theory [4], which also derives geometric structure from type-theoretic primitives, though the methods and targets differ.

## 2 The Distinction Axiom

**Definition 2.1** (Distinction)**.** A *distinction* is a record consisting of:

- a carrier type $S$ : Set,
- two elements $\ell, r : S$,
- a separation proof $\ell \neq r$,
- a coverage function $cover : (x : S) \to (x \equiv \ell) \uplus (x \equiv r)$.

In Agda:

```
record Distinction : Set₁ where
  field
    S : Set
    l : S
    r : S
    l≠r : l ≠r
    cover : (x : S) →(x ≡l) ⊎(x ≡r)
```

Coverage is the critical field. Without it, $S$ could contain hidden elements unreachable by $\ell$ and $r$, and the classification that follows would be incomplete. With it, every element is provably one of two values, and every function $S \to S$ is determined by its action on $\ell$ and $r$.

## 3 Four-Case Endomorphism Classification

**Theorem 3.1** (Classification). *Let $d$ be a distinction. Every endomorphism $f : S \to S$ is classified by one of exactly four cases:*

$$EndoCase \;=\; constL \mid constR \mid id \mid dual$$

*The classification is:*

- ***sound**: $interpret(classify(f))f$,*
- ***injective**: $interpret(c)interpret(c') \;\Rightarrow\; c \equiv c'$,*
- ***unique**: $interpret(c)f \;\Rightarrow\; c \equiv classify(f)$.*

*Proof sketch.* Classification dispatches on $cover(f(\ell))$ and $cover(f(r))$, yielding $2 \times 2 = 4$ branches. Soundness follows by case analysis and symmetry of equality. Injectivity is proved by a $4\times4$ exhaustion: for each pair of distinct cases, the assumption $interpret(c)interpret(c')$ produces a contradiction via $\ell \neq r$. Uniqueness combines soundness and injectivity. All three are verified by Agda in both volumes. □ □

The key consequence: the endomorphism monoid of any distinction is *exactly* four elements. Not three (which would orphan a case), not five (which would require a third boundary value).

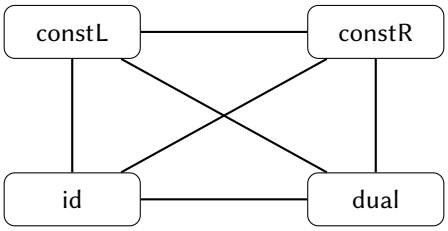


Figure 1: The four endomorphism cases and their pairwise distinctness. Every pair is connected because every pair is provably $\neq$. This is $K_4$.

## 4 Drift Hierarchy and Acyclicity

Classification applies not only within a single distinction but *across universe levels.* Applying the classification functor to its own output produces a tower of drift states, one per level of the type hierarchy.

**Definition 4.1** (Drift State)**.** A drift state packages a distinction at a given universe level together with its derived structure. The step function lifts a drift state one level higher.

**Definition 4.2** (Reachability)**.** Strict reachability $s \prec^+ t$ is the transitive closure of the step relation. Reflexive reachability $s \preceq t$ adds the identity.

**Theorem 4.3** (Ranking forces acyclicity)**.** *If there exists a rank function $rank : DriftState \to N$ such that each step strictly increases the rank, then the drift hierarchy is acyclic: no state is strictly reachable from itself.*

*Proof sketch.* A cycle $s \prec^+ s$ forces $suc(rank(s)) \leq rank(s)$ by monotonicity, which is impossible in $N$. Verified as `law11-1-ranked-acyclic` in the Agda source. □ □

## 5 Presentation Fixpoint

**Definition 5.1** (Presentation)**.** A presentation of the endomorphism classification is a type $X$ together with maps $interpret : X \to (S \to S)$ and $classify : (S \to S) \to X$ satisfying soundness and injectivity.

**Theorem 5.2** (Presentation collapse)**.** *Every presentation is isomorphic to $EndoCase$. In particular, any admissible observable on proofs factors through the canonical four-element base.*

This fixpoint property means that no matter how the four-element classification is re-encoded or re-presented, the underlying structure is invariant. There is no presentation freedom.

## 6 $K_4$ as Unique Survivor

**Definition 6.1** (Graph)**.** A graph is a type $V$ of vertices with a symmetric, irreflexive edge relation $Edge : V \to V \to \mathsf{Set}$.

**Definition 6.2** (Canonical $K_4$)**.** The canonical $K_4$ has $V = EndoCase$ and $Edge(a, b) = (a \neq b)$.

Since all four endomorphism cases are mutually distinct (Section 3), every pair is connected. The result is the complete graph on four vertices.

**Theorem 6.3** (Presentation uniqueness)**.** *Every $K_4$ graph presentation—any alternative vertex type $X$ equipped with a bijection to $EndoCase$—produces a graph isomorphic to the canonical $K_4$.*

*Proof sketch.* The bijection transports the edge relation (inequality) faithfully. Distinctness is invariant under renaming. Verified as `law13-1-presentation-iso` in the source. □ □

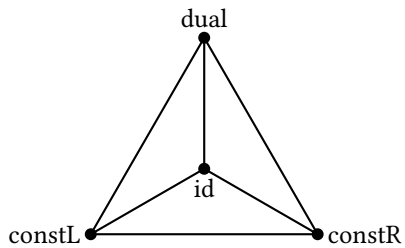


Figure 2: $K_4$: the complete graph on the four endomorphism cases.

## 7 The Elimination Chain

The full derivation is summarized in Table 1.

| # | Survivor | Why it survives | Eliminated |
|---|---|---|---|
| 1 | Distinction ($D_0$) | Binary separation with coverage. | Vacuous carriers, partial classifications. |
| 2 | Four endomorphisms | $2 \times 2 = 4$ cases, mutually distinct. | $K_1$, $K_2$, $K_3$ (orphan cases). |
| 3 | Drift acyclicity | Ranking forces well-foundedness. | Cyclic hierarchies, infinite descent. |
| 4 | Presentation fixpoint | Observables factor through canonical base. | Non-canonical representations. |
| 5 | $K_4$ | Unique complete graph on four vertices. | All proper subgraphs and supergraphs. |

Table 1: The eliminative chain from distinction to $K_4$.

# 8 The Layered Architecture

The formal chain is realized by two Agda files compiled under the same flags. *Void* is self-contained. *Form* imports *Void* and builds the physical identification layer on top of it.

| | ***Void* (Foundation)** | ***Form* (Identification)** |
|---|---|---|
| File | `Void.lagda.tex` | `Form.lagda.tex` |
| Lines | $\sim$25,000 | $\sim$9,000 |
| Module | `Void` | `Form` |
| Imports | `Agda.Primitive` only | `open import Void` |
| Central verb | Eliminate | Identify |
| Content | $D_0 \to K_4 \to$ invariants $\to$ arithmetic $\to$ spectrum | $K_4$ invariants $\to$ physical constants |
| Equality | Propositional (Dec $(x \equiv y)$) | Inherits from Void |
| Flags | `--safe --without-K` | `--safe --without-K` |
| Postulates | 0 | 0 |
| Result | $(V, E, d, \chi, \kappa) = (4, 6, 3, 2, 8)$ | 35-field `StandardModelFromK4` |

The separation is clean: *Void* proves the existence and uniqueness of $K_4$ and derives all arithmetic from it. *Form* imports those results and maps them onto measured quantities. No new axiom enters at the identification layer. Every number that appears in *Form* was already forced in *Void*; *Form* only names the correspondence.

# 9 Structural Theorems

The following identifiers are machine-verified in both volumes. Each row is a `refl` or an absurdity witness $\lambda\,()$ closed by the type checker.

| Agda identifier | Content |
|---|---|
| `Two-distinction` | $D_0$ is explicitly constructed; not just irrefutable |
| `EndoCase` | Exactly 4 endomorphisms; 4 constructors, none can be added |
| `law13-1-presentation-iso` | Any $K_4$ presentation is isomorphic to the canonical one |
| `K4PhysRep-singleton` | The physics record has exactly one inhabitant |
| `theorem-discrete-continuum-map-unique` | The bridge fractions $11/72$ and $11/576$ are the unique irreducible survivors |
| `physics-presupposes-distinction` | `K4PhysRep` $\to D_0$: the loop closes |
| `measurement-outcome-distinction` | Any binary measurement contains $D_0$ |
| `NormalizedScaleChoice` | $c = \hbar = G = 1$ forced; $\forall S : a = b = m = 1$ |

## 10 Addressed Objections

### "The $K_4$ uniqueness only holds within your self-built system."

The isomorphism proof `law13-1-presentation-iso` shows that any alternative presentation of the four-case classification—any type $X$ with its own interpretation, classification, soundness, and injectivity—is isomorphic to `EndoCase`. Alternative presentations do not survive the round-trip. The argument is a universal statement over all presentations satisfying the same typed interface.

### "How do you know the two volumes do not share a hidden assumption?"

*Form* explicitly imports *Void* via `open import Void`. The shared assumption is therefore transparent, not hidden: it is Agda's core type theory under `--safe --without-K`, plus everything *Void* derives from it. *Form* adds no postulates and no new axioms. It identifies *Void*'s forced invariants with measured constants—a mapping, not a derivation.

### "No dynamics means no physical theory."

Correct. This paper is not a physical theory. It is a classification theorem: a single binary distinction forces $K_4$ through a chain in type theory. The numerical consequences of $K_4$'s invariants, and their comparison with measured constants, are treated separately in the physics companion [3]. The formal chain stands or falls independently of any physical interpretation.

## 11 Formal Environment

Both volumes are single literate Agda files compiled under:

```
{-# OPTIONS --safe --without-K #-}
```

No standard library. No postulates. No axiom K. All infrastructure—equality, negation, decidability, coproducts, natural numbers, ordering—is built from scratch in *Void*. *Form* inherits these definitions via `open import Void`.

The `--safe` flag forbids `postulate`, `TERMINATING`, and other escape hatches. The `--without-K` flag forbids Streicher's axiom K, ensuring compatibility with homotopy type theory.

Verification:

```
git clone https://github.com/de-johannes/Void-and-Form
cd Void-and-Form
agda --safe --without-K Void.lagda.tex # Foundation
```

```
agda --safe --without-K Form.lagda.tex # Identification layer
```

*Form* depends on *Void*; compile *Void* first. If both files type-check, every theorem stated in this paper is verified.

## 12 Discussion

The result is a purely combinatorial classification theorem: the only graph that survives the self-consistency filter forced by a single distinction is $K_4$. The derivation introduces no free parameter and makes no appeal to geometry, physics, or model-theoretic semantics.

That the result is reached by two independent formal developments strengthens the claim. A structure found only by elimination might depend on the eliminative framing. A structure found only by construction might depend on the constructive framing. A structure found by both depends on neither—it depends on the logic alone.

Several questions remain open:

- Can the exhaustive classification be sharpened to a single categorical universal property?
- What is the precise relationship between the drift hierarchy and ordinal notations in proof theory?
- Does the presentation fixpoint admit a formulation in terms of initial algebras or terminal coalgebras?
- Can the result be reproduced in alternative proof assistants (Lean 4, Coq) to confirm independence from Agda-specific features?

## 13 Conclusion

A single binary distinction—a type with two provably unequal, covering elements—forces, through a chain verified in Agda under `--safe --without-K`, exactly four mutually distinct endomorphisms, a well-founded drift hierarchy, a canonical presentation fixpoint, and the complete graph $K_4$ as the unique survivor.

*Void* ($\sim$25,000 lines) derives the chain and proves the invariant record to be a singleton. *Form* ($\sim$9,000 lines) imports the result and identifies the invariants with measured physical constants. Both compile under the same flags with zero postulates. The formal content is settled. The physical interpretation is open.

## References


[1] J. M. Wielsch, *Void: The Eliminative Derivation of $K_4$ from Distinction*, literate Agda (`Void.lagda.tex`), $\sim$25 000 lines, April 2026. `DOI: 10.5281/zenodo.17826218`.

[2] J. M. Wielsch, *Form: The Identification of $K_4$ Invariants with Physics*, literate Agda (`Form.lagda.tex`), $\sim$9 000 lines, April 2026. Imports `Void`.

[3] J. M. Wielsch, *Computed Invariants of $K_4$: A Physics Record*, companion paper (numerical results), April 2026.

[4] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013.